
QR-Iteration-komplexe-Eigenwerte

$$\begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 1 \\ -1 & -1 & 1 \end{bmatrix}$$

Exakte Eigenwerte

$$\begin{bmatrix} 1 \\ 1 - \sqrt{3}i \\ 1 + \sqrt{3}i \end{bmatrix}$$
$$\begin{bmatrix} 1.0 \\ 1.0 - 1.73205080756888i \\ 1.0 + 1.73205080756888i \end{bmatrix}$$

QR Iteration (ohne Shift)

```
array([[ 1.          ,  1.41421356, -0.81649658],  
       [-1.41421356,  1.          , -0.57735027],  
       [ 0.81649658,  0.57735027,  1.          ]])
```

```
array([[ 1.          ,  1.63299316,  0.49236596],  
       [-1.63299316,  1.          ,  0.30151134],  
       [-0.49236596, -0.30151134,  1.          ]])
```

```
array([[ 1.          ,  1.70560573, -0.26010244],  
       [-1.70560573,  1.          , -0.15249857],  
       [ 0.26010244,  0.15249857,  1.          ]])
```

```
array([[ 1.          ,  1.72532437,  0.13193885],  
       [-1.72532437,  1.          ,  0.07647191],  
       [-0.13193885, -0.07647191,  1.          ]])
```

```
array([[ 1.          ,  1.73036183, -0.06621046],  
       [-1.73036183,  1.          , -0.03826394],  
       [ 0.06621046,  0.03826394,  1.          ]])
```

```
array([[ 1.          ,  1.7316281 ,  0.03313552],  
       [-1.7316281 ,  1.          ,  0.01913547],  
       [-0.03313552, -0.01913547,  1.          ]])
```

```
array([[ 1.          ,  1.7319451 , -0.01657155],  
       [-1.7319451 ,  1.          , -0.00956817],  
       [ 0.01657155,  0.00956817,  1.          ]])
```

```
array([[ 1.          ,  1.73202438,  0.00828625],  
       [-1.73202438,  1.          ,  0.00478414],  
       [-0.00828625, -0.00478414,  1.          ]])
```

```
array([[ 1.          ,  1.7320442 , -0.00414318],
       [-1.7320442 ,  1.          , -0.00239208],
       [ 0.00414318,  0.00239208,  1.          ]])
```

```
array([[ 1.00000000e+00,  1.73204916e+00,  2.07159943e-03],
       [-1.73204916e+00,  1.00000000e+00,  1.19603963e-03],
       [-2.07159943e-03, -1.19603963e-03,  1.00000000e+00]])
```

Die Grenzmatrix ist Block-diagonal.

Der erste Block ist

```
array([[ 1.          ,  1.73204916],
       [-1.73204916,  1.          ]])
```

mit Eigenwerten

```
array([1.+1.73204916j, 1.-1.73204916j])
```

Der zweite Block ist

```
array([[1.]])
```

mit Eigenwert

```
array([1.])
```